Monetary Policy Transmission and Building Land Scarcity: A Panel Local Projections Analysis of Regional Austrian Housing Markets

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Abstract

Acknowledgements

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4. Data and Descriptive Evidence

The empirical analysis rests on a balanced quarterly panel of Austrian districts covering the period Q1 2017 to Q4 2025. The political district is the spatial unit. The panel

therefore comprises 116 districts observed over 36 quarters, which yields 4 176 district-quarter observations. Vienna is retained at the level of its individual municipal districts, so that the Viennese housing market enters the analysis at the same granularity as the remaining political districts rather than as a single aggregate. The following subsections describe the outcome variable, the monetary policy shock, the supply-constraint measure, and the control variables in turn. Further, the subsections document descriptive evidence that motivates the following empirical analysis, before the closing subsection documents how these components are combined into the estimation panel along with preliminary diagnostics of the data.

4.1. Real Estate Price Data

Residential real estate prices are provided by "DataScience Service GmbH" (2026), which compiles district-level price measures from transaction data. The data distinguish three property categories, average prices for single-family houses (EFH), average prices for condominium apartments (ETW), and building land prices, each expressed in € per m². "DataScience Service GmbH" (2026) data is based on land-register transaction records ("Grundbuchauszüge"). Building land is not treated as a housing outcome and is therefore excluded from the price series used in estimation. Over the full sample the price data are complete, with an observation for every district in every quarter and no missing values in any of the three categories.

The primary outcome is a district-level composite price constructed as a fixed-weight average of the house and apartment series. For district i in quarter t, the composite price level is defined as

$$p_{i,t}^{\text{comp}} = w_i^{\text{house}} \, \text{EFH}_{i,t} + w_i^{\text{apt}} \, \text{ETW}_{i,t}, \tag{4.1}$$

where $\text{EFH}_{i,t}$ and $\text{ETW}_{i,t}$ are the average house and apartment prices in € per m², and the weights w_i^{house} and w_i^{apt} are held constant over the sample and are derived from the district housing stock, as described in the next paragraph. The underlying price data is in nominal terms, as time fixed-effects will absorb national inflation in the estimation. Composing the series in levels and taking logarithms afterwards avoids time-varying endogenous weights, while the fixed weights allow each district to be represented by

the property mix that actually characterizes its market. The outcome transformation is explained in detail in section 5.4 . This choice motivates the property-specific robustness outcomes discussed in Section 6, where the house and apartment series are examined separately.

The composite weights are based on the building stock recorded in the 2011 Statistik Austria building and housing census, which reports the number of dwellings by the number of dwellings within the building on the district-level (Statistik Austria, 2026c). Let D_i^1, D_i^2, $D_i^{3\text{–}5}$, $D_i^{6\text{–}10}$, $D_i^{11\text{–}20}$, and D_i^{21+} denote the 2011 dwelling counts in district i across the six building-size classes. Dwellings in single-dwelling and two-dwelling buildings are assigned to the house category, the latter to capture semi-detached and duplex dwellings that behave like the owner-occupied house market, while dwellings in buildings with three or more dwellings are assigned to the apartment category. The house weight is then the house-category share of the district dwelling stock,

$$w_i^{\text{house}} = \frac{D_i^1 + D_i^2}{D_i^1 + D_i^2 + D_i^{3\text{–}5} + D_i^{6\text{–}10} + D_i^{11\text{–}20} + D_i^{21+}}, \qquad w_i^{\text{apt}} = 1 - w_i^{\text{house}}. \tag{4.2}$$

The weights are fixed at their 2011 values so that they predate the estimation window and remain exogenous to price developments during the sample, following the same logic that governs the supply-constraint measure below. Because the weights are time-invariant, cross-district composition is held fixed and only the underlying prices vary over time.

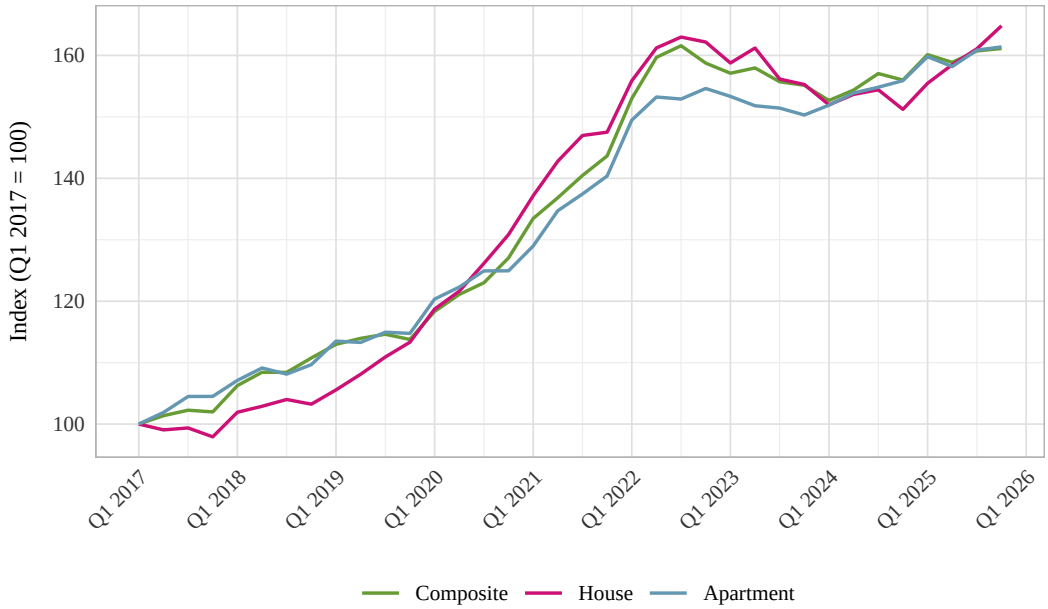


Figure 1: District price indices by property type
Unweighted district medians, Q1 2017 = 100.

Across the sample, the composite price averages roughly € 3 260 per m², with a standard deviation of about € 1 590 and a range that spans from under € 900 to over € 13 000 per m². This wide dispersion is the first indication of substantial regional heterogeneity in Austrian housing markets. The house and apartment series differ markedly in level, with average house prices around € 4 200 per m² exceeding average apartment prices of about € 2 870 per m². Both series are right-skewed, so that a small number of high-price districts sit well above the median. The evidence presented is descriptive and is not intended to support causal claims, since the differences reported can reflect urbanity, dwelling composition, income, and other district characteristics rather than the transmission of monetary policy. The purpose is to establish that the data contain economically meaningful regional heterogeneity in price levels, price dynamics, and supply conditions, and to surface the features that shape the specification and interpretation of the local projections.

The price dispersion is closely related to supply conditions. Grouping districts by the supply-constraint classification (see 4.3), the mean composite price in inelastic districts is roughly € 4 410 per m² against about € 2 480 in elastic districts, so that the more supply-constrained districts operate in a distinctly higher price regime. As emphasized

above, this level difference is not evidence of differential monetary transmission, since inelastic districts are also more urban and more apartment-heavy. It does, however, confirm that the supply classification separates districts that differ systematically in their housing markets, which is a precondition for the interaction analysis that follows.

Quarterly composite price growth averages about 1.3% over the sample, with elastic districts growing marginally faster on average than inelastic districts. More important for the empirical strategy is the time-series behavior of price growth. Pooling the within-district changes, the autocorrelation of quarterly composite growth is positive and persistent, at about 0.23 at the first lag and remaining close to 0.20 through the fourth lag. A positive autocorrelation means that a quarter of above-average price growth tends to be followed by further above-average growth, so that shocks to prices are not reversed immediately but propagate over several quarters. This gradual adjustment, commonly termed price momentum, is a well-documented feature of housing markets and reflects the slow diffusion of information, search frictions, and the staggered nature of transactions in illiquid property markets.

The persistence has a direct methodological implication. If lagged price changes predict current growth and are also correlated with the timing of monetary policy surprises, omitting them would confound the estimated response with the ongoing momentum of the local market. Following Aastveit and Anundsen (2018), who include four lags of house-price changes in their panel local projections precisely to absorb this momentum, the control vector therefore includes four lagged quarterly changes of the composite price. The observed lag profile, which stays economically meaningful out to the fourth lag, supports the choice of four lags rather than a shorter window. The same persistence, combined with the overlapping cumulative outcomes of the local projections, also suggests the serial-correlation concern that governs the treatment of standard errors discussed in Section 5.

4.2. Monetary Policy Shocks

The monetary policy shock is constructed from the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019), which records high-frequency changes in a range of asset prices within tight windows around ECB policy announcements. The chosen instrument is the surprise in the twenty-year overnight index swap rate (OIS

20Y). This maturity is selected because residential real estate prices depend on expected long-term financing conditions, so that surprises at the long end of the yield curve are the component of monetary policy most relevant for housing valuations. The twenty-year and thirty-year German government bond yield surprises are retained only as alternative instruments for robustness.

The event-level surprises are aggregated to the quarterly frequency of the panel by summing all surprises within a quarter into a net quarterly shock. Alongside the net measure, the sum of positive surprises, the sum of negative surprises, and the sum of absolute surprises are recorded for each quarter, which supports the signed decomposition used to examine asymmetric responses. The sign convention follows the interpretation of the underlying instrument, a negative OIS 20Y surprise denotes a downward revision of long-term rate expectations and therefore a monetary easing surprise. This convention matters for the interpretation of the estimated coefficients and is carried through consistently in Section 5. Over the sample the mean net quarterly surprise is negative, at roughly -0.59, consistent with the prevailing monetary easing stance during much of the period, and the largest absolute net surprises occur in Q3 2019 and around the onset of the tightening cycle in 2022 and 2023.

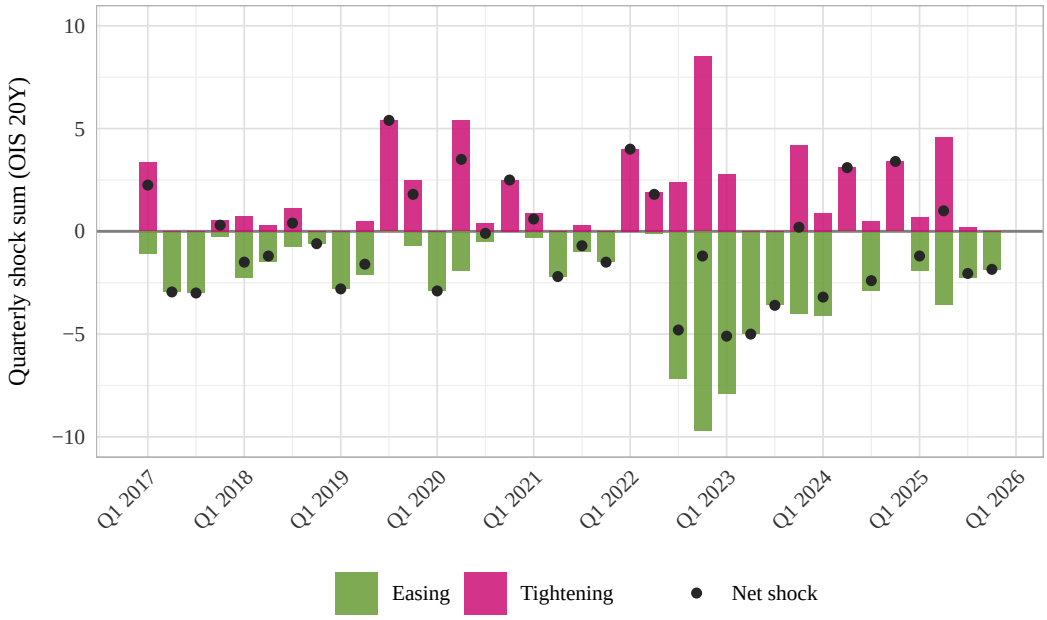


Figure 2: Quarterly monetary policy shock components
Easing surprises denoted as negative, tightening surprises denoted as positive; points = net quarterly sum

A preliminary diagnostic examines how far the quarterly net sum conceals offsetting surprises within a quarter. The cancellation ratio, defined as the net shock (net sum of shocks in a quarter) divided by the absolute sum of surprises in a quarter, has a mean of about 0.69, which indicates that easing and tightening surprises frequently occur within the same quarter and partly offset in the net measure. The ratio is a diagnostic rather than an additional regressor, and the extent of within-quarter cancellation motivates the alternative shock aggregations, such as the easing-only and tightening-only components, that are examined as robustness checks.

The quarterly net OIS 20Y surprise is predominantly negative over the sample, in line with the accommodative stance that characterized much of the period, before turning sharply positive around the onset of the tightening cycle in Q3 2022. Both easing and tightening quarters are well represented, with an easing component present in 31 of the 36 quarters and a tightening component in 26, and both signs occurring within the same quarter in 21 quarters. This overlap is the source of the within-quarter cancellation documented above, and the presence of both directions of surprise is what makes the sample suitable for examining whether price responses are asymmetric across monetary

policy phases. Because the shock is common to all districts within a quarter, its aggregate effect is absorbed by the time fixed effects, and the descriptive variation reported here bears on the identifying interaction only through its differential incidence across the supply groups.

4.3. Supply-Constraint Measure

Regional differences in housing supply are captured through the district building land reserve recorded in the ÖROK monitoring of building land reserves (ÖROK Monitoring of Building Land Reserves, 2025), which measures the share of vacant developable land in total designated building land. A lower reserve indicates less room for additional construction and therefore a more supply-inelastic district. To predate the estimation window, the 2014 level of the reserve is used throughout, so that the classification cannot respond to price developments during the sample.

Districts are classified as supply-inelastic using the twenty-fifth percentile of the reserve distribution outside Vienna, a threshold of around 22%, which is then applied to all districts. This procedure classifies 47 districts as inelastic and 69 as elastic. Of the inelastic districts, 24 lie outside Vienna and 23 are Viennese municipal districts, so that all Viennese districts and all Salzburg districts fall into the inelastic group. The measure is a proxy for supply conditions rather than a directly estimated supply elasticity, and it is strongly structured along geographic and compositional lines. In particular, the reserve measure correlates at about 0.91 with the fixed house weight, so that inelastic districts tend also to be apartment-heavy. *HIER NOCH EINEN ABSATZ ZU INVESTOR BEHAVIOR EINFÜGEN*. District fixed effects absorb time-invariant level differences of this kind, but the entanglement of supply constraints with geography and property composition remains a concern for interpretation, which is why property-specific outcomes, a Vienna-excluded specification, and a continuous reserve measure are reported as robustness checks in Section 6.

The building land reserve varies considerably across districts, with a median of about 24% and values ranging from close to zero to above 45%. The supply classification is strongly clustered by federal state. All Viennese districts and all Salzburg districts fall into the inelastic group, while Burgenland contains no inelastic districts, and the median reserve in Vienna, at under 5%, is far below that of the predominantly rural

states. This geographic concentration is economically intuitive, since the most supply-constrained districts are the dense urban and Alpine areas where developable land is genuinely scarce, but it also implies that the supply measure is entangled with region and with property composition.

This entanglement carries two implications that are taken up in the empirical analysis. First, district fixed effects remove time-invariant differences in price levels and composition, but they do not by themselves rule out that districts differing in these correlated characteristics also respond differently to a common shock for reasons other than supply elasticity. Second, because the interaction with the shock is by construction non-zero only for inelastic districts, and because all Viennese districts are inelastic, the identifying variation is concentrated in a structurally distinctive set of markets. These considerations motivate the property-specific outcomes, the Vienna-excluded specification, and the continuous reserve measure that are reported as robustness checks, and they caution against reading the baseline interaction as a clean measure of supply elasticity alone.

4.4. Control Variables

The control set comprises district-level population, the unemployment rate, and net income per capita, each aligned to the quarterly frequency of the panel. Population is available quarterly from the central residence register and enters directly (Statistik Austria, 2026b). Unemployment is available annually at the district level and is carried across the four quarters of each year (Statistik Austria, 2026a). Net income per capita is constructed by dividing annual district net income totals (Statistik Austria, 2026d) by the corresponding population, so that within-year variation in income per capita partly reflects population movements. Following, this creates mechanical correlation between income growth and population growth. Since these series are annual in origin, the quarterly panel repeats the annual figures within each year, and this construction is reflected in the way the control lags behave.

Several series require construction at the tail of the sample, where source data are not yet available. Annual net income is observed through 2022 and is extrapolated for 2023 to 2025 using the nominal Tariflohnindex (OeNB - Österreichische Nationalbank, 2026), so that all districts share a common wage-index growth path over the extrapolated years. District unemployment is observed through 2024 and carried forward to 2025, and

population is carried forward from the first quarter of 2025 to the remaining quarters of that year. These constructed values are complete observations, but they do not represent newly observed district information. Their share is largest at the end of the sample and declines at longer horizons as later quarters drop out of the estimation window. To ensure that these choices do not drive the findings, the analysis reports a sensitivity sample that excludes the constructed tail.

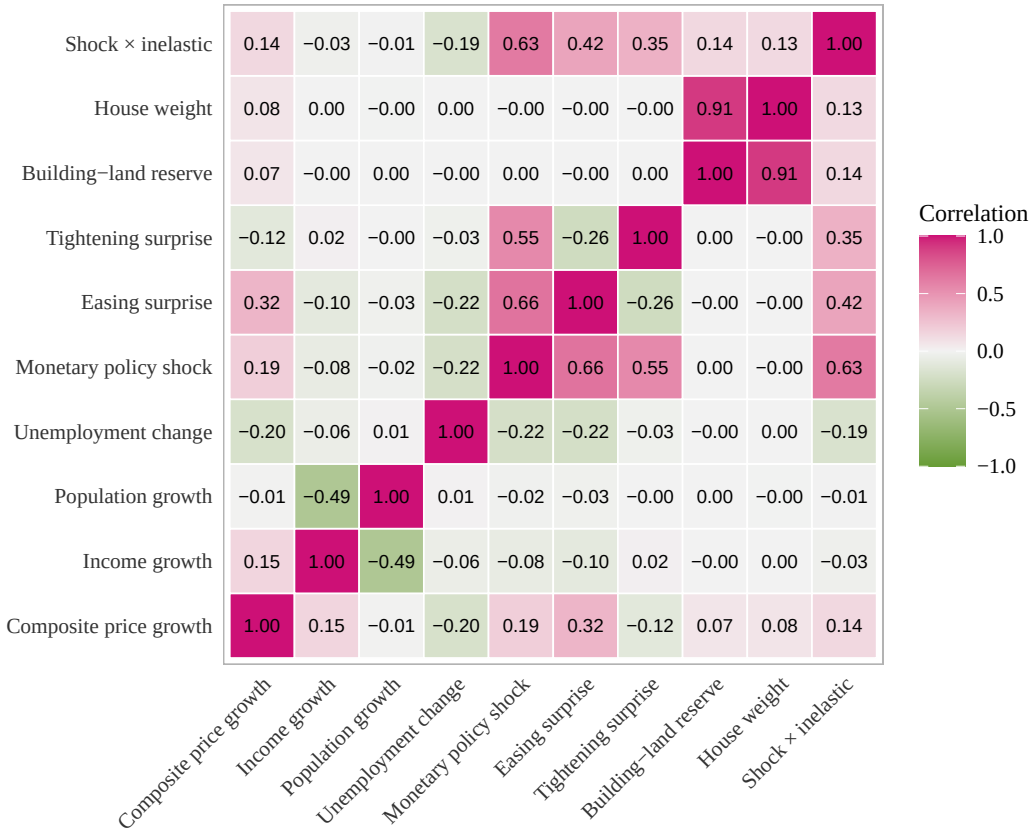


Figure 3: Pairwise correlations of key variables

Figure 3 reports the pairwise correlations among the outcome, the controls, the shock and its signed components, the supply measures, and the shock-by-inelastic interaction. The matrix serves two purposes, it documents that the shock behaves as an external

regressor with respect to district characteristics, and it surfaces one strong association that shapes the interpretation of the results.

The most reassuring result for identification is the relation of the shock to the district-level variables. The shock is essentially uncorrelated with the two time-invariant supply measures, with a coefficient at 0.00 with both the building land reserve and the fixed house weight. Further, it correlates only weakly with the district controls, at -0.08 with income growth, -0.02 with population growth, and -0.22 with the change in unemployment. The near-zero correlation with the time-invariant measures is mechanical rather than coincidental, since a common national surprise carries no cross-sectional variation that could line up with fixed district traits, but it nonetheless confirms that the identifying variation in the shock does not proxy for the supply classification. The modest negative correlation with the unemployment change is consistent with monetary easing coinciding with weaker labor market conditions, and it motivates the inclusion of the controls rather than raising a collinearity concern.

The shock also relates to the outcome in the expected direction (see 5.3). The net shock correlates at 0.19 with quarterly composite price growth, where the signed decomposition is more informative, with the easing component correlating at 0.32 and the tightening component at -0.12. Under the convention that easing surprises enter negatively, these signs indicate that price growth is higher in quarters with larger easing surprises, which is the unconditional counterpart of the response the local projections estimate conditionally. Within the shock block, the net measure correlates at 0.66 with the easing component and at 0.55 with the tightening component, while the two signed components correlate at -0.26 with each other, reflecting the within-quarter cancellation documented in Section 4.2. The interaction term correlates at 0.63 with the net shock, as expected since it is the shock scaled by the inelastic indicator, and only weakly, at around 0.13 to 0.14, with the two supply measures.

Among the controls, the strongest association is the -0.49 correlation between income growth and population growth. This is largely mechanical, as income per capita is constructed by dividing an annual district income total by quarterly population, so that movements in the denominator enter both series with opposite sign, as noted in Section 4.4. The remaining control correlations are small, with composite price growth correlating at -0.20 with the change in unemployment and at 0.15 with income growth, which is consistent with local demand conditions without indicating redundancy among

the regressors.

The one association that requires closer attention is the 0.91 correlation between the building land reserve and the fixed house weight. Because inelastic districts are predominantly urban and apartment-heavy, the supply classification and the composite property mix carry closely related cross-sectional information, so that the inelastic indicator partly captures the difference between apartment markets and house markets rather than land scarcity alone. This entanglement is the central measurement concern of the analysis, and it is not resolved by the correlation matrix itself.

HIER NOCH EINEN ABSATZ ZU INVESTOR BEHAVIOR EINFÜGEN (step 5): acknowledge that apartments carry more weight in urban and inelastic districts, but argue with literature that investor-dominated apartment markets react more strongly to monetary shocks than owner-occupier house markets, framing this as a related channel that reinforces rather than confounds the supply story, with appropriate caveats.

Two variables considered at the design stage are not included. Tourism intensity is not available at the district level on a consistent basis over the sample period, since it is released by the federal states under differing conventions and frequencies, and it is therefore recorded as a data limitation rather than approximated with inconsistent series. Net migration, which features in the international literature (Aastveit & Anundsen, 2018), could be incorporated later should consistent annual district data become available, and population growth serves as the closest available proxy for local demand pressure in the meantime.

4.5. Panel Construction and Preliminary Diagnostics

The estimation panel is assembled by merging the price series, the composite weights, the supply-constraint classification, the quarterly shock, and the three control variables on the district and quarter keys. The price series and controls vary by district and quarter, the shock is common across districts within a quarter, and the supply classification and composite weights are time-invariant district characteristics. Districts are identified by a consistent numeric code across all sources, and the Viennese municipal districts are aligned to the three-digit district scheme used throughout the panel.

A set of preliminary diagnostics confirms that the merge preserves the intended struc-

ture of the data. All component inputs join without duplicate or unmatched keys, every district appears in every quarter, and the price series contain no non-positive or non-finite values, so the merged panel is balanced and complete at 4 176 district-quarter observations (covering 116 districts over 36 quarters). Summary statistics for the outcome, the shock, the supply measure, and the controls, together with the integrity checks underlying these statements, are reported in Appendix 1. The descriptive evidence discussed above draws on this panel, and the diagnostics reported here establish that the regularities documented in the preceding subsections reflect the data.

The dependent variable and the lagged controls introduce structural missing values by construction rather than through gaps in the source data. The cumulative outcome (as described in 5) at horizon h is measured relative to the quarter preceding the shock, and four lagged quarterly changes enter as controls, so the usable sample declines from 3 596 observations at the shortest horizon to 2 204 at the longest. All 116 districts remain represented at every horizon, but each additional horizon removes the most recent quarter, so that comparisons of coefficients across horizons also compare progressively earlier samples. This feature is inherent to the local projection design and is disclosed alongside the estimates.

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A. Appendix

Variable	N	Mean	SD	Min	Median	Max
Outcome						
Composite price (€/m^2)	4 176	3 261	1 593	868	2 876	13 500
House price (€/m^2)	4 176	4 203	2 699	849	3 200	16 212
Apartment price (€/m^2)	4 176	2 865	1 539	742	2 471	13 515
Composite price growth (log change)	4 060	0.0135	0.0238	-0.1080	0.0118	0.1570
Monetary policy shock						
Net shock (OIS 20Y)	36	-0.59	2.66	-5.10	-1.20	5.40
Easing component	36	-2.29	2.30	-9.70	-1.90	0.00
Tightening component	36	1.70	2.06	0.00	0.72	8.50
Supply measure						
Building land reserve (%)	116	21.2	10.2	0.1	23.8	45.5
House weight	116	0.500	0.273	0.003	0.580	0.874
Controls						
Population	4 176	77 446	47 622	1 900	65 370	305 314
Unemployment rate (%)	4 176	6.0	3.3	1.6	5.0	17.9
Net income per capita (€)	4 176	22 680	4 797	5 155	21 680	51 092

Note. N differs by variable block. Prices and controls are observed at the district-quarter level. The monetary policy shock varies only over the 36 quarters. The supply measure and the fixed dwelling-stock weights are time-invariant, they vary over the 116 districts. Composite price growth is a quarterly log change and is undefined in the first quarter of each district, which accounts for its lower count.

Table 1: Summary statistics of the analysis variables

Diagnostic	Result
Balanced panel (districts × quarters)	116 × 36 = 4 176
Unique district-quarter keys	4 176 (no duplicates)
Missing or unexpected keys	0
Districts observed in all 36 quarters	116
Source inputs joined without key errors	7 of 7
Non-positive or non-finite price cells	0
Weight-sum and composite identity errors	0

Table 2: Panel construction and integrity diagnostics